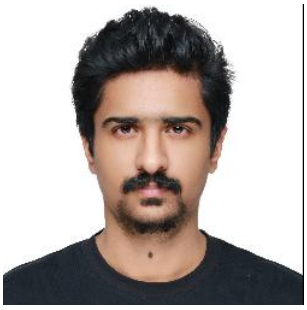


Dr. ADIL HUSSAIN

PhD in *Traffic Information Engineering & Control*.

EV Charging Optimization, Demand Forecasting & Dynamic Pricing, Reinforcement Learning, Unmanned Systems, AI for Intelligent Transportation.

Nationality: Pakistani | Permanent **Location:** Rawalpindi, Pakistan | Current **Location:** Xi'an, China

Phone: +86 18629407811 (CHN) | +92 3025264036 (PAK)

Email: adilgujjar598@gmail.com | 18629407811@163.com

Links: Website | ORCID | Google Scholar | Scopus | Web of Science | ResearchGate | LinkedIn |

SUMMARY

Transportation-AI researcher with a Ph.D. in Traffic Information Engineering and Control (Chang'an University, June 2026), developing transformer-based and reinforcement-learning methods for EV charging demand forecasting, dynamic pricing, and charging-station optimization. Author of 24 journal articles, 15 conference papers, and 7 book chapters, with 450+ citations, an i10 index of 16, and an h-index of 12 (Google Scholar). Secondary research includes UAVs, autonomous systems, and predictive modelling. I served as a reviewer for 16 journals and a technical program committee member for IEEE BTS-I2C (2024-2026) and IEEE ICATEI (2025). Seeking a postdoctoral or faculty position in Artificial Intelligence, Transportation Engineering or Autonomous systems.

EDUCATION

- **Ph.D., *Traffic Information Engineering and Control*** **Sep 2022 – Jun 2026**

Chang'an University, Xi'an, China

Thesis Title: Research on Integrated Framework and Strategies for Electric Vehicle Charging Stations Optimization

Supervisor: Professor QingChang Lu

- **Master of Science in *Computer Science* (MSCS)** **Sep 2019 – Jan 2022**

Bahria University, Islamabad, Pakistan

Supervisor: Dr. Kashif Naseer Qureshi

- **Bachelor of Science in *Information Technology* (BSIT)** **Sep 2014 – Jul 2018**

PMAS Arid Agriculture University, Rawalpindi, Pakistan

RESEARCH & PROFESSIONAL EXPERIENCE

- **Teaching Assistant (Tutor)** **Sep 2024 – Jun 2026**

Academic English Writing Workshop by International Education School, Chang'an University

Delivered lectures and workshops on research methodology and academic writing to help international graduate students; assisted lead tutors and mentored new researchers in early research skills; supported students in preparing journal and conference submissions. I was recognized as Outstanding Teaching Assistant (2024) for my contribution, with acknowledgements to the workshop in published articles.

- **Research Assistant *(MS Research Fellow)*** **Sep 2019 – Sep 2021**

Cyber Reconnaissance and Combat Lab (CRC), Islamabad, Pakistan

Conducted research in security-related topics, especially in smart home systems and networks, and co-authored 1 paper.

- **Android Developer** **Mar 2022 – Nov 2022**

Speckpro Digital, Islamabad, Pakistan

Built Android front ends and integrated REST APIs for client apps including "Salon Wizz," the e-commerce app "Kasa Beeston," and the "CASA Tenant Management System"; one-month trial converted to a full-time role.

- **Junior Android Developer & Intern** **Feb 2019 – Sep 2019**

O3 Interfaces (Pvt) Ltd., Islamabad, Pakistan

Worked on the "UBL Digital" mobile banking app for features (bill payments, card management, digital onboarding) and improved UI/UX features; 3-month internship converted to a full-time role. Implemented and tested REST APIs, performed QA and daily bug fixes,

TEACHING & MENTORING

- **Teaching:** Academic English Writing Workshop, International Education School, Chang'an University (Sep 2024 – Jun 2026): lectures on research methodology and scientific writing for international graduate students.
- **Mentoring:** guided early-stage researchers in preparing journal and conference submissions.
- **Recognition:** Outstanding Teaching Assistant (2024).
- **Teaching interests:** Artificial Intelligence, Intelligent Transportation Systems, Transportation Data Analytics, Research Methodology.

SELECTED PUBLICATIONS

Total: 24 journal articles, 15 conference papers, 7 book chapters

Google Scholar: 450+ citations, i10-index 16, h-index 12

Refer to **APPENDIX** for complete list of publications.

** Corresponding author; ** co-corresponding author.*

1. **Adil Hussain**, Qing-Chang Lu*, Kashif Naseer Qureshi. *Context-Aware Temporal Fusion Transformer for Short-Term EV Charging Demand Prediction*. **Sustainable Energy, Grids and Networks** (2026)

   (Web of Science 2025: SCIE, Q1, IF: 5.7)

2. **Adil Hussain**, Qing-Chang Lu*, Kashif Naseer Qureshi**, Khalid Javeed. *On-Demand Dynamic Charging Pricing for Electric Vehicles*. **Egyptian Informatics Journal** (2026).

   (Web of Science 2025: SCIE, Q2, IF: 4.2)

3. **Adil Hussain**, Qing-Chang Lu*, Sanam Shahla Rizvi, Shixin Wang, Se Jin Kwon**. *Short Term Demand Forecasting of Electric Vehicle Charging Stations using Context Aware Temporal Transformer Model*. **Scientific Reports** (2025)

   (Web of Science 2025: SCIE, Q1, IF: 4.9)

4. **Adil Hussain*,** Vishwanath Eswarakrishnan**, Ayesha Aslam, Sajib Tripura. *Charging stations demand forecasting using LSTM based hybrid transformer model*. **Scientific Reports** (2025) (Web of Science 2025: SCIE, Q1, IF: 4.9)

5. Abdullahi, Muhammad Rabiu, Qing-Chang Lu, **Adil Hussain***, Sajib Tripura, Peng-Cheng Xu, ShiXin Wang. *Location optimization of EV charging stations: A custom K-means cluster algorithm approach*. **Energy Reports** (2024)

   (Web of Science 2025: SCIE, Q2, IF: 6.6)

6. **Adil Hussain***, Ayesha Aslam. *A Reinforcement Learning based Adaptive Dynamic Pricing for EV Charging Stations*. **World Transport Congress (WTC)**, 2026, Tianjin, China.

7. **Adil Hussain***, Ayesha Aslam. *A Cost-Aware Optimal Charging Time Algorithm for EVs in a Parking Lot*. **World Transport Congress (WTC)** 2024, Qingdao, China.

8. **Adil Hussain**, Vineet Dhanawat, Ayesha Aslam, Faizan Zaman. *Electricity Load Forecasting Using Attention-Based Hybrid Deep Learning Model*. **IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAIET)**, 2024, Malaysia.

9. **Adil Hussain**, Ayesha Aslam, Rundong Wang, Kashif Naseer Qureshi. *Advanced Learning Techniques for AI in Renewable Energy (Chapter 7).* ***Artificial Intelligence-based Renewable Energy Systems Standards, Communication Systems, and Data Networks*** 1st Edition, Elsevier, eBook ISBN: 9780443406195.

HONOURS, AWARDS & FUNDING

- **2026 Excellent International Graduate Student** **Jun 2026**

International Education School, Chang'an University, Xi'an, China

- **2025 Star of Study Award** **Dec 2025**

International Education School, Chang'an University, Xi'an, China

- **2025 Excellent International Student** **Dec 2025**

International Education School, Chang'an University, Xi'an, China

- **2024 Outstanding Teaching Assistant** **May 2025**

International Education School, Chang'an University, Xi'an, China

- **2024 Star of Study Award** **Dec 2024**

International Education School, Chang'an University, Xi'an, China

- **One Belt, One Road Scholarship by the Chinese Government** **Sep 2022 - Jun 2026**

One Belt, One Road Scholarship for Doctoral Studies at Chang'an University, Xi'an, China.

SKILLS & EXPERTISE

- **Research:** Scientific writing, literature review, research design, academic publishing, data analysis, dissertation supervision, teaching
- **Technical:** Machine/deep learning (transformers, attention-based models, LSTM, CNN, graph neural networks), reinforcement learning, time-series forecasting, predictive modelling.
- **Programming:** Python, Java, C#, C++
- **Research Tools:** LaTeX, Origin, QGIS, Git/GitHub, Jupyter, Google Colab, PyCharm; SQL databases.
- **Reference Management:** EndNote, Mendeley, Zotero
- **Languages:** English (Excellent) · Chinese (Basic) · Punjabi (Native) · Urdu (Excellent)

JOURNALS, CONFERENCES, SEMINARS & REVIEWER

- **Journals served as a Reviewer**
  - **IEEE Journals:** IEEE Transactions on Smart Grid
  - **Elsevier Journals:** Electric Power Systems Research, Journal of Engineering Research
  - **Springer and Wiley Journals:** Scientific Reports, Discover Artificial Intelligence, Journal of Big Data, International Journal of Machine Learning and Cybernetics, Knowledge and Information Systems. Cluster Computing. **Wiley:** Expert Systems

- **Frontiers Journals:** Frontiers in Artificial Intelligence, Frontiers in Computer Science
- **MDPI:** Sustainability, World Electric Vehicle Journal, Energies, Future Transportation

- **Conferences served as a Technical Program Committee (TPC) & Reviewer**
  - 3rd IEEE BTS-I2C 2026, *Malaysia*
  - 2nd IEEE BTS-I2C 2025, *Malaysia*
  - 1st IEEE International Conference on Advanced Technologies in Energy and Informatics (ICATEI) 2025, *Indonesia*
  - 1st IEEE Beyond Technology Summit on Informatics Int'l Conference (BTS-I2C) 2024, *Malaysia*

REFERENCES

Professor Qing-Chang Lu *(Ph.D. Supervisor)*

Chang'an Scholar Chair Professor, Vice Dean / School of Electronics and Control Engineering

Department Chair / Department of Traffic Information and Control Engineering, Chang'an University.

Email: qclu@chd.edu.cn

***New Affiliation:** School of Management, Shanghai University, Shanghai, China.

Email: qclu@shu.edu.cn

Dr. Kashif Naseer Qureshi *(Master's Supervisor)*

Associate Professor, Department of Computer Science, Bahria University, Islamabad, Pakistan.

Email: knqureshi@gmail.com

***New Affiliation:** Department of Electronics and Computer Engineering, University of Limerick, Limerick, Ireland.

Email: kashifnaseer.qureshi@ul.ie

Professor Haiping Liu

Head of the Academic English Writing Workshop

Organized by the International Education School (IES), Chang'an University, Xi'an, China.

Email: lhpholly@chd.edu.cn

Professor Shuguang Li

School of Electronics and Control Engineering, Department of Traffic Information and Control Engineering, Chang'an University.

Email: shgli@chd.edu.cn

APPENDIX: COMPLETE PUBLICATION LIST

List of all journal articles, conference papers, and book chapters

Journal Articles

2026

1. **Adil Hussain**, Qing-Chang Lu*, Kashif Naseer Qureshi. *Context-Aware Temporal Fusion Transformer for Short-Term EV Charging Demand Prediction*. **Sustainable Energy, Grids and Networks** (2026)
2. **Adil Hussain**, Qing-Chang Lu*, Kashif Naseer Qureshi*, Khalid Javeed. *On-Demand Dynamic Charging Pricing for Electric Vehicles*. **Egyptian Informatics Journal** (2026).
3. Tariq, Zhu Wei, Kamran Shaukat*, Ayesha Aslam, **Adil Hussain**, Md. Mahboob Alam**. *A Spatiotemporal Forecast-Guided SAC-Learning Framework for 3D UAV-BS Trajectory Planning*. **Scientific Reports**
4. Sajib Tripura, Qing-Chang Lu*, Sanam Shahla Rizvi, **Adil Hussain**, Tae-Sun Chung**. *Graph Attention Network-Based Accident Risk Aware Message Dissemination for IoV Safety*. **Expert Systems with Applications**
5. Mostak Ahamed, Ibrahim Kholilullah, Hairong Gu*, Md Moin Shikder, Munju Mia, Sajib Tripura, **Adil Hussain**. *Automated control integration of solar and wind energy for sustainable DC microgrid deployment in desert environments for EV charging*. **Discover Electronics**

2025

1. **Adil Hussain**, Qing-Chang Lu*, Sanam Shahla Rizvi, Shixin Wang, Se Jin Kwon*. *Short Term Demand Forecasting of Electric Vehicle Charging Stations using Context Aware Temporal Transformer Model*. **Scientific Reports** (2025)
2. **Adil Hussain*,** Vishwanath Eswarakrishnan, Ayesha Aslam, Sajib Tripura. *Charging stations demand forecasting using LSTM based hybrid transformer model*. **Scientific Reports** (2025)
3. Qing-Chang Lu*, Shixin Wang, Peng-Cheng Xu, Jing Li, Xu Meng, **Adil Hussain**. *Modeling the Dependency Relationship of Coupled Power and Transportation Network*. **Energy**
4. Ayesha Aslam, Yang Xiaojun*, Kashif Naseer Qureshi**, **Adil Hussain**, Kayhan Zrar Ghafoor. *GST-Former: GCN-based Spatial Temporal Transformer Model for Interaction-Aware Vehicle Trajectory Prediction*. **Journal of Engineering Research**
5. Faizan Zaman, Zhigang Xu*, **Adil Hussain,** Anees Ullah, Khalid Zaman *Enhancing Driver Emotion Recognition Through Deep Ensemble Classification*. **Journal of Intelligent and Connected Vehicles**
6. Sajib Tripura, Qing-Chang Lu*, **Adil Hussain**, Tanjim Mehmud. *Optimizing traffic safety message dissemination and resource allocation using adaptive deep reinforcement learning in fog-enabled internet of vehicles network*. **Egyptian Informatics Journal**

7. Sajib Tripura, Qing-Chang Lu*, Dhonita Tripura, MD Ibrahim Kholilullah, Arunav Malik Avi, Mostak Ahmed, **Adil Hussain**. *Adaptive Q-Learning-Based Event-Prioritized QoS and Incentive Optimization for Enhancing Safety in Vehicular Fog Networks*. **Egyptian Informatics Journal**

8. Amna Ikram*, Sunnia Ikram, El-Sayed M. El-Kenawy, **Adil Hussain**, Amal H. Alharbi*, Marwa M. Eid. *A fuzzy-optimized hybrid ensemble model for yield prediction in maize-soybean intercropping system*. **Frontiers in Plant Science**

2024

9. Abdullahi, Muhammad Rabiu, Qing-Chang Lu, **Adil Hussain***, Sajib Tripura, Peng-Cheng Xu, ShiXin Wang. *Location optimization of EV charging stations: A custom K-means cluster algorithm approach*. **Energy Reports** (2024)

10. Tariq, Vishwanath Eswarakrishnan*, **Adil Hussain**, Zhu Wei, Muhammad Uzair. *A Novel Energy Replenishment Algorithm to Increase the Network Performance of Rechargeable Wireless Sensor Networks*. **Sensors**

11. **Adil Hussain***, Ayesha Aslam, Sajib Tripura, Vineet Dhanawat, Varun Shinde. *Weather Forecasting Using Machine Learning Techniques: Rainfall and Temperature Analysis*. **Journal of Advances in Information Technology**

12. **Adil Hussain***, Ayesha Aslam. *Hate Speech against Women and Immigrants: A Comparative Analysis of Machine Learning and Text Embedding Techniques*. **Journal of Applied Research and Technology**

13. **Adil Hussain***, Ayesha Aslam. *Ensemble-based Approach using Inception V2, VGG-16, and Xception Convolutional Neural Networks for Surface Cracks Detection*. **Journal of Applied Research and Technology**

14. **Adil Hussain**, Amna Khatoon*, Ayesha Aslam, Tariq, Muhammad Asif Khosa. *A Comparative Performance Analysis of Machine Learning Models for Intrusion Detection Classification*. **Journal of Cyber Security**

15. Ayesha Aslam, **Adil Hussain***. *A Performance Analysis of Machine Learning Techniques for Credit Card Fraud Detection*. **Journal of Artificial Intelligence**

16. **Adil Hussain***, Ayesha Aslam. *Cardiovascular Disease Prediction Using Risk Factors: A Comparative Performance Analysis of Machine Learning Models*. **Journal of Artificial Intelligence**

2023

17. **Adil Hussain**, Kashif Naseer Qureshi*, Khalid Javeed, Musaed Alhussein. *Enhanced Intelligent Intrusion Detection System for E-Commerce*. **Computer Systems Science & Engineering**

18. **Adil Hussain**, Asad Ullah*, Ayesha Aslam, Amna Khatoon. *A Modified Siamese Network for Facial Assimilation*. **WSEAS Transactions on Signal Processing**

2022

19. Kashif Naseer Qureshi, Adi Alhudhaif, **Adil Hussain**, Saleem Iqbal, Gwanggil Jeon. *Trust aware energy management system for smart homes*. **Computers & Electrical Engineering**

Conference Papers

2026

1. **Adil Hussain***, Ayesha Aslam. *A Reinforcement Learning-Based Adaptive Dynamic Pricing for EV Charging Stations*. **World Transport Congress (WTC)**, Tianjin, China.
2. Sajib Tripura, Noman Iqbal, Zahirul Islam Kajal, **Adil Hussain**, Ayesha Aslam, Mostak Ahamed. *Traffic Speed Prediction using LSTM and Spatial GCN-based Feature Fusion.* **IEEE 3rd INTERNATIONAL CONFERENCE ON COMPUTING, APPLICATIONS AND SYSTEMS** Bangladesh (2026) - ACCEPTED

2025

1. Sajib Tripura, **Adil Hussain**, Mostak Ahamed, Dhonita Tripura, Ayesha Aslam. *A Dual Attention Mechanism-Based Enhanced Residual Network for Skin Cancer Detection*. **IEEE 2nd International Conference on Computing, Applications, and Systems (COMPAS)**, Bangladesh.
2. Ayesha Aslam, Sajib Tripura, Dhonita Tripura, Mostak Ahamed, **Adil Hussain**. *Households Electricity Load Forecasting using Deep Learning Techniques: An Efficacy Analysis of Attention Mechanism for Hourly Forecasting*. **IEEE 2nd International Conference on Computing, Applications, and Systems (COMPAS)** Bangladesh.
3. **Adil Hussain**, Muhammad Yasir, Noman Iqbal, Sajib Tripura, Ayesha Aslam. *Heart Disease Classification Using Machine Learning Techniques: An Impact Analysis of GridSearchCV-based Optimization*. **IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA)** Yemen.
4. Ayesha Aslam, **Adil Hussain**, Noman Iqbal, Muhammad Yasir, Sajib Tripura. *Solar Panel Dust Detection: Efficacy Analysis and Optimization using Deep Learning.* **IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA)** Yemen.
5. Peng-Cheng Xu, Qing-Chang Lu, Feng Tao, Jing Li, **Adil Hussain**, Wang Rundong. *Network-Based Resilience Assessment of Metro Stations: The Relation of Infrastructures and Passengers.* **104th Transportation Research Board (TRB) Annual Meeting**, United States.

2024

6. **Adil Hussain***, Ayesha Aslam. *A Cost-Aware Optimal Charging Time Algorithm for EVs in a Parking Lot*. **World Transport Congress (WTC)** Qingdao, China.
7. **Adil Hussain**, Vineet Dhanawat, Ayesha Aslam, Noman Iqbal, Sajib Tripura. *Credit Card Fraud Detection Using Machine Learning Techniques: Dealing with Imbalanced Data Using*

Over-Sampling and Under-Sampling Methods. **IEEE Beyond Technology Summit on Informatics International Conference (BTS-I2C)** Indonesia.

8. **Adil Hussain**, Kashif Naseer Qureshi, Raja Waseem Anwar, Ayesha Aslam. *A Novel SCD11 CNN Model Performance Evaluation with Inception V3, VGG16 and ResNet50 Using Surface Crack Dataset*. **IEEE 2nd International Conference on Unmanned Vehicle Systems-Oman (UVS)** Oman.
9. **Adil Hussain**, Kashif Naseer Qureshi, Faizan Zaman, Ayesha Aslam. *Minor Surface Cracks Detection using SCD11 Convolutional Neural Network*. **IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA)** Yemen.
10. **Adil Hussain**, Kashif Naseer Qureshi, Ayesha Aslam, Tariq, Muhammad Rabiu Abdullahi. *TSC18 Convolutional Neural Network for Traffic Sign Classification*. **IEEE International Conference on Emerging Smart Technologies and Applications (eSmarTA)** Yemen.
11. **Adil Hussain**, Vineet Dhanawat, Ayesha Aslam, Faizan Zaman. *Electricity Load Forecasting Using Attention-Based Hybrid Deep Learning Model*. **IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAIET)** Malaysia.
12. Amna Khatoon, Weixing Wang, Asad Ullah, Ishfaq Ahmad, **Adil Hussain**, Mardeni Roslee. *Advancement in Pavement Condition Assessment: An AI-Based Computer Vision Approach*. **IEEE Multimedia University Engineering Conference (MECON)** Malaysia.
13. Ayesha Aslam, Sajib Tripura, Muhammad Rabiu Abdullahi, Vineet Dhanawat, **Adil Hussain**. *Traffic Sign Classification Using German Traffic Sign Dataset Leveraging Vision Transformer (ViT) Model*. **3rd International Conference on Data Science and Emerging Technologies (DaSet), Springer** Malaysia.
14. Faizan Zaman, Zhigang Xu, **Adil Hussain**, Ayesha Aslam, Muhammad Rabiu Abdullahi. *Concrete and Pavement Cracks Detection Leveraging Vision Transformer (ViT) Model*. **IEEE International Conference on Electrical and Information Technology (IEIT)** Indonesia.

2023

15. **Adil Hussain**, Ayesha Aslam, Tariq. *Traffic Sign Classification Using a Novel TSC18 Convolutional Neural Network*. **9th International Conference of Sabaragamuwa University of Sri Lanka.** (Abstract) Sri Lanka.

Book Chapters

2026

1. **Adil Hussain**, Ayesha Aslam, Mahad Ahmad. *Artificial Intelligence in Agriculture: Applications, Challenges, and Future Directions (Chapter 3).* ***AI-Driven Farming Systems With Minimal Human Intervention*****,** 1st Edition: **IGI-Global**, eISBN13: 9798337392974.

2025

2. **Adil Hussain**, Ayesha Aslam, Rundong Wang, Kashif Naseer Qureshi. *Advanced Learning Techniques for AI in Renewable Energy (Chapter 7).* ***Artificial Intelligence-based Renewable Energy Systems Standards, Communication Systems, and Data Networks,*** 1st Edition, **Elsevier**, eBook ISBN: 9780443406195.

3. Ayesha Aslam, **Adil Hussain**, Kashif Naseer Qureshi. *Big Data Analytics for AI in Renewable Energy Networks (Chapter 9).* ***Artificial Intelligence-based Renewable Energy Systems Standards, Communication Systems, and Data Networks,*** 1st Edition, **Elsevier**, eBook ISBN: 9780443406195.

4. **Adil Hussain**, Kashif Naseer Qureshi, Ayesha Aslam. *Novel Machine and Deep Learning and Training Techniques for AI-based Healthcare Systems (Chapter 7).* ***Artificial Intelligence-based Smart Healthcare Systems: New Standards, Technologies and Communication Systems,*** 1st Edition, **Elsevier**, Paperback ISBN: 9780443264764.

5. Ayesha Aslam, Kashif Naseer Qureshi, **Adil Hussain**. *Advanced AI-Based Healthcare Systems Applications and Services (Chapter 4).* ***Next Generation AI Language Models in Research: Promising Perspectives and Valid Concerns,*** 1st Edition. **Taylor & Francis - CRC Press**, eBook ISBN: 9781032667911.

6. **Adil Hussain**, Peng-Cheng Xu, Wang Shixin, Kashif Naseer Qureshi. *Advanced AI-Based Healthcare Systems Applications and Services (Chapter 6).* ***Next Generation AI Language Models in Research: Promising Perspectives and Valid Concerns,*** 1st Edition. **Taylor & Francis - CRC Press**, eBook ISBN: 9781032667911.

2024

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7. **Adil Hussain**, Kashif Naseer Qureshi. *Standards and Policies Adoption for AIoT Networks (Chapter 4).* ***Artificial Intelligence of Things (AIoT): New Standards, Technologies and Communication Systems*****,** 1st Edition, **Taylor & Francis - CRC Press**, eBook ISBN: 9781003430018.